

1. Data Loading & Initial Inspection

```
import numpy as np

import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt


data = pd.read_csv("global_cancer_patients_2015_2024.csv")

data.info()

data.duplicated().sum()
```

What I Did

- Imported core Python data libraries
- Loaded CSV dataset
- Checked data structure
- Checked duplicate records

Why This Step?

Industry workflow always starts with:

1. Data structure understanding
2. Data type validation
3. Duplicate & null detection

Business Angle

Before analysis, analysts must validate data integrity to avoid misleading conclusions.

2. Age Distribution Analysis

Charts Used:

1. **KDE Plot** (Density Plot)
2. **Histogram**

```
sns.kdeplot(data["Age"], fill=True)
```

```
sns.histplot(data["Age"], bins=30)
```

Chart Types:

- KDE → Continuous density estimation
- Histogram → Frequency distribution

Why These Charts?

- Age is a **continuous variable**
- Histogram shows distribution count
- KDE shows smooth distribution trend

Statistical Summary:

```
data["Age"].describe()
```

What to Explain:

- Mean → Average age
- Std → Variability
- IQR → Middle 50% spread

Insight:

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Range: 20 to 89 years

Mean Age: 54.42 years

Standard Deviation: 20.22

Interquartile Range (IQR): 37 (Q1) to 72 (Q3)

This suggests a broad representation of both young and elderly patients in the dataset, which supports age-based comparative analysis.

3. Gender Distribution

Chart Used:

Bar Chart

```
data["Gender"].value_counts()
```

```
sns.barplot(...)
```

Why Bar Chart?

- Gender is **categorical variable**
- Bar chart best for category comparison

Interpretation:

Male has highest count → balanced gender representation.

Business Angle:

The dataset contains three gender categories(Male, Female ,other) with the most common being Male (16,796 records).

Gender distribution is sufficient for evaluating gender-specific survival trends and severity outcomes.

4. Country Distribution

Chart Used:

Pie Chart

```
plt.pie(...)
```

Why Pie Chart?

- Shows proportion of total dataset
- Useful for percentage share visualization

Interpretation:

Data comes from 10 countries with similar representation.

Business Angle:

Patients come from 10 different countries/regions, with Australia being the most represented (5,092 patients). Number of data points for each country is almost same.

This diversity enables cross-country comparison of cancer outcomes and treatment economic.

5. Cancer Type Distribution

Chart Used:

Bar Chart

```
sns.barplot(...)
```

Why?

- Cancer Type = categorical variable
- Bar chart ideal for frequency comparison

Interpretation:

8 cancer types, roughly balanced dataset.

Business Angle:

We have in total 8 types of cancer, with each cancer having approx same number of data points under the label, most common cancers are colon cancer followed by Prostate cancer.

Task

6. Cancer Stage Distribution

Chart Used:

Bar Chart

```
sns.barplot(...)
```

Why?

- Stage is ordinal categorical variable (0–4)
- Bar chart helps visualize stage dominance

Interpretation:

Stage 2 most common.

Stages almost equally distributed.

Business Meaning:

Cancer Stage have 5 stages with values ranging from 0 to 4 , with stage 2 the most common one , and each stage have almost same number of data points under its label.

Task

7. Treatment Cost Distribution

Charts Used:

1. KDE Plot
2. Histogram

```
sns.kdeplot(data["Treatment_Cost_USD"], fill=True)
```

```
sns.histplot(data["Treatment_Cost_USD"], bins=30)
```

Why?

Treatment cost is continuous variable.

What to Check:

```
data["Treatment_Cost_USD"].describe()
```

```
data["Treatment_Cost_USD"].skew()
```

Interpretation:

If skew $\approx 0 \rightarrow$ symmetric distribution
No heavy skew $\rightarrow$ balanced cost distribution.

Business Insight:

Treatment cost USD have no skewness , and there are almost same number of data points under each bin as observed by histogram.

8. Risk Factor Summary Statistics

```
summary = data[column_of_interest].agg(['mean', 'std', 'min', 'max'])  
  
column_of_interest= ['Genetic_Risk', 'Air_Pollution', 'Alcohol_Use',  
 'Smoking', 'Obesity_Level']
```

What I Did:

Computed central tendency & dispersion for:

- Genetic Risk
- Air Pollution
- Alcohol Use
- Smoking
- Obesity Level

Why?

To check if variables are standardized.

Interpretation:

Similar means & std → likely standardized scoring scale.

Business Meaning:

These variables have nearly identical means and standard deviations, indicating they were likely designed on the same standardized scale. They are essential in studying interaction effects (e.g., genetic risk × smoking) on survival.

9. Risk Factors vs Severity (Regression Analysis)

Chart Used:

Regression Plot (Scatter + Regression Line)

```
sns.regplot(x=x, y=y)
```

Why Regression Plot?

- Risk factors are continuous
- Severity score is continuous
- Want to examine linear relationship

Statistical Model:

Used:

```
linregress()
```


It gives:

- Slope
- Intercept
- R-value
- R^2
- p-value

What to Explain:

- **Slope** → Direction of impact
- **R^2** → Strength of relationship (0 to 1)
- Higher R^2 = stronger predictor

Business Insight:

Identifies strongest lifestyle/genetic predictor of cancer severity.